

Washington based Center for Strategic and International Studies (CSIS) reports that the following cyber-attacks took place in recent months.

February 2021

- In a unit of Oxford University working on the Covid-19 vaccine, there was an incident of unauthorised access to information
- North Korea based hackers extorted billions of cryptocurrencies

March 2021

- A group of hackers used Facebook to broadcast malicious and suspicious links to capture information
- Computer Emergency Response Team (CERT) of India received evidence of Chinese hackers spying in sensitive sectors

Tools and libraries for fingerprinting Android apps

Before covering the tools and libraries, we should first be clear about the backend or base of any Android app. An Android package (APK) file is the base installation file of a classical Android App. The hackers or sniffers try to corrupt this file, and then send it via special URLs or links so that the receivers download and install it on their Android smartphones.

Before installation of any Android App, its fingerprinting and analysis is necessary. There are a number of sources and repositories where such APK files are available for download. Figure 2 gives a few cloud repositories where thousands of Android apps are available. These should be analysed first before direct installation in a phone.

Many smartphone users download games and apps from the cloud without checking whether they are secure or not.

A number of tools and libraries are available for the fingerprinting and forensic evaluation of Android apps and the associated APK files. Google Play store also uses a number

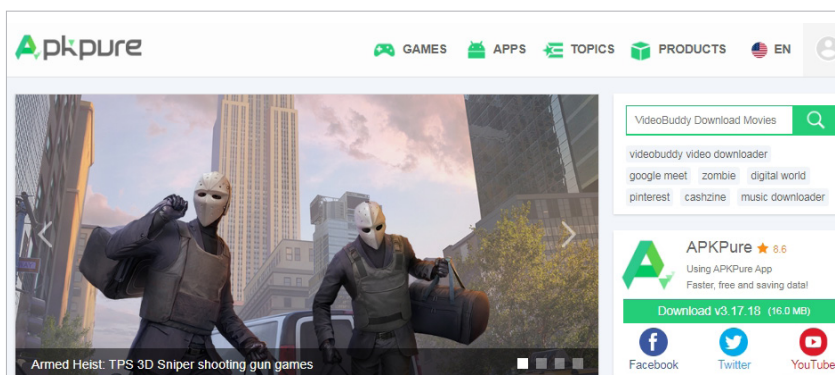


Figure 2: Cloud based repository of Android APKs

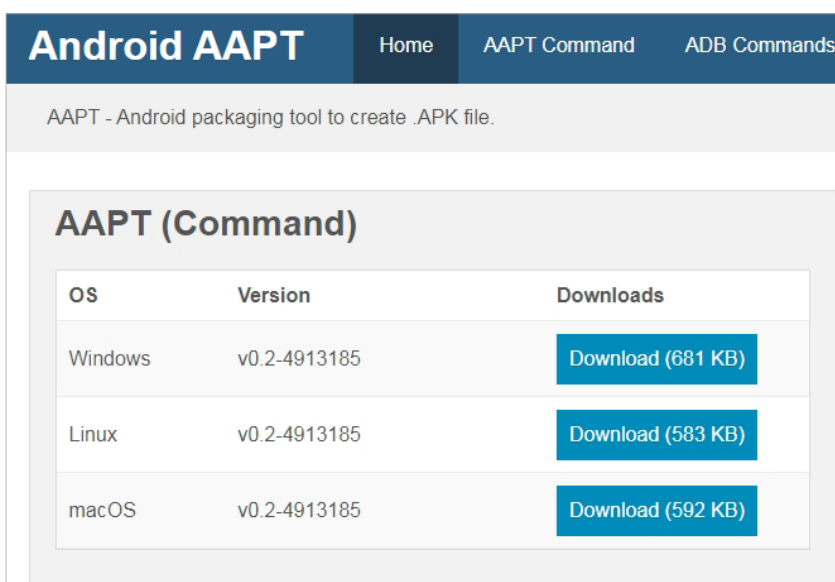


Figure 3: Downloading the Android AAPT tool

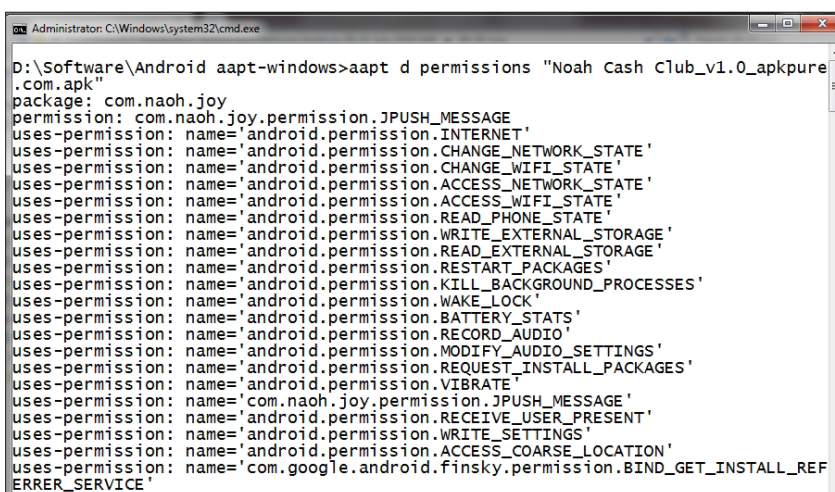


Figure 4: Display permissions of Android app in AAPT tool